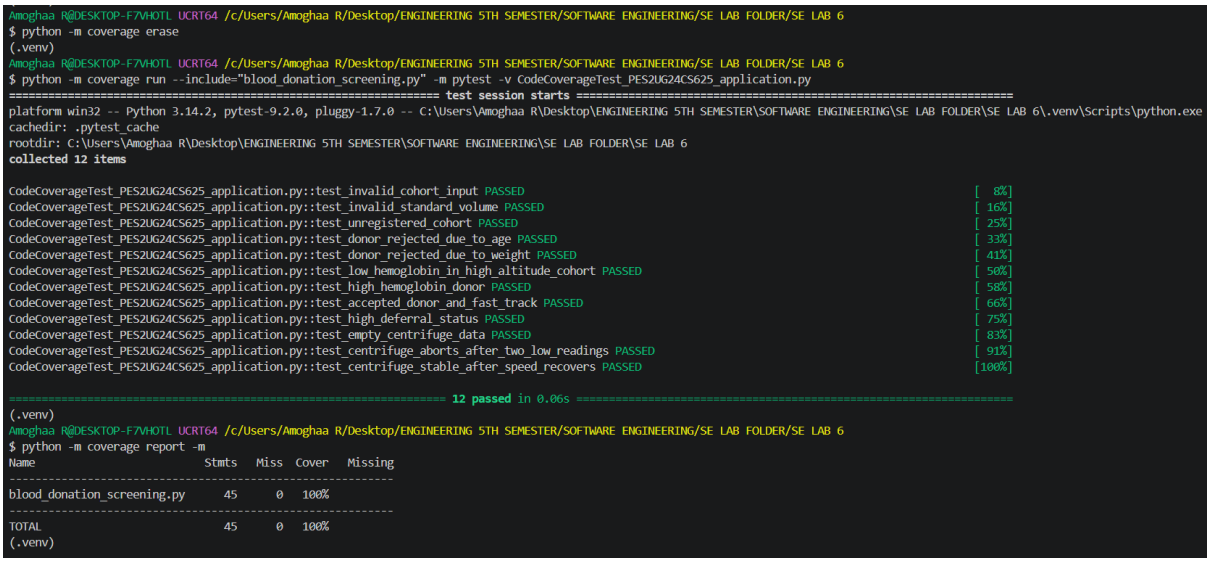
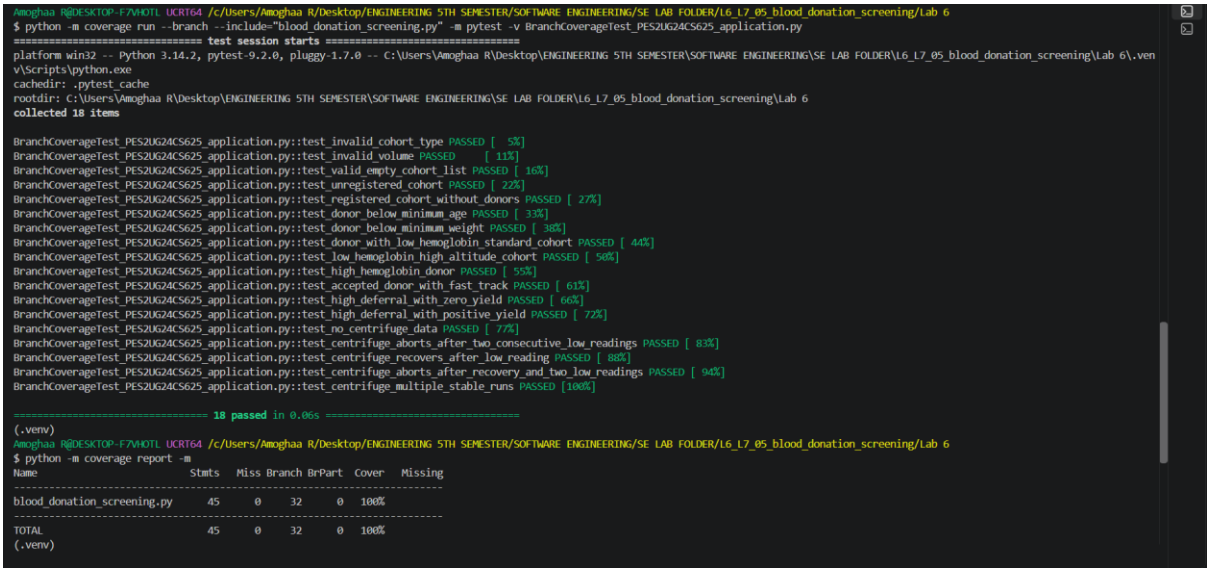
1. Coverage (Code and Branch) analysis (Terminal Output Screenshots)



```
Amoghaa R@DESKTOP-F7VHOTL UCRT64 /c/Users/Amoghaa R/Desktop/ENGINEERING 5TH SEMESTER/SOFTWARE ENGINEERING/SE LAB FOLDER/SE LAB 6
$ python -m coverage erase
(.venv)
Amoghaa R@DESKTOP-F7VHOTL UCRT64 /c/Users/Amoghaa R/Desktop/ENGINEERING 5TH SEMESTER/SOFTWARE ENGINEERING/SE LAB FOLDER/SE LAB 6
$ python -m coverage run --include="blood_donation_screening.py" -m pytest -v CodeCoverageTest_PES2UG24CS625_application.py
============================= test session starts ==============================
platform win32 -- Python 3.14.2, pytest-9.2.0, pluggy-1.7.0 -- C:\Users\Amoghaa R\Desktop\ENGINEERING 5TH SEMESTER\SOFTWARE ENGINEERING\SE LAB FOLDER\SE LAB 6\.venv\Scripts\python.exe
cachedir: .pytest_cache
rootdir: C:\Users\Amoghaa R\Desktop\ENGINEERING 5TH SEMESTER\SOFTWARE ENGINEERING\SE LAB FOLDER\SE LAB 6
collected 12 items

CodeCoverageTest_PES2UG24CS625_application.py::test_invalid_cohort_input PASSED                                    [  8%]
CodeCoverageTest_PES2UG24CS625_application.py::test_invalid_standard_volume PASSED                                 [ 16%]
CodeCoverageTest_PES2UG24CS625_application.py::test_unregistered_cohort PASSED                                     [ 25%]
CodeCoverageTest_PES2UG24CS625_application.py::test_donor_rejected_due_to_age PASSED                               [ 33%]
CodeCoverageTest_PES2UG24CS625_application.py::test_donor_rejected_due_to_weight PASSED                            [ 41%]
CodeCoverageTest_PES2UG24CS625_application.py::test_low_hemoglobin_in_high_altitude_cohort PASSED                  [ 50%]
CodeCoverageTest_PES2UG24CS625_application.py::test_high_hemoglobin_donor PASSED                                   [ 58%]
CodeCoverageTest_PES2UG24CS625_application.py::test_accepted_donor_and_fast_track PASSED                           [ 66%]
CodeCoverageTest_PES2UG24CS625_application.py::test_high_deferral_status PASSED                                    [ 75%]
CodeCoverageTest_PES2UG24CS625_application.py::test_empty_centrifuge_data PASSED                                   [ 83%]
CodeCoverageTest_PES2UG24CS625_application.py::test_centrifuge_aborts_after_two_low_readings PASSED                [ 91%]
CodeCoverageTest_PES2UG24CS625_application.py::test_centrifuge_stable_after_speed_recovers PASSED                  [100%]

============================== 12 passed in 0.06s ==============================
(.venv)
Amoghaa R@DESKTOP-F7VHOTL UCRT64 /c/Users/Amoghaa R/Desktop/ENGINEERING 5TH SEMESTER/SOFTWARE ENGINEERING/SE LAB FOLDER/SE LAB 6
$ python -m coverage report -m
Name                          Stmts   Miss  Cover   Missing
-----------------------------------------------------------
blood_donation_screening.py      45      0   100%
-----------------------------------------------------------
TOTAL                            45      0   100%
(.venv)
```



```
Amoghaa R@DESKTOP-F7VHOTL UCRT64 /c/Users/Amoghaa R/Desktop/ENGINEERING 5TH SEMESTER/SOFTWARE ENGINEERING/SE LAB FOLDER/L6_L7_05_blood_donation_screening/Lab 6
$ python -m coverage run --branch --include="blood_donation_screening.py" -m pytest -v BranchCoverageTest_PES2UG24CS625_application.py
============================= test session starts ==============================
platform win32 -- Python 3.14.2, pytest-9.2.0, pluggy-1.7.0 -- C:\Users\Amoghaa R\Desktop\ENGINEERING 5TH SEMESTER\SOFTWARE ENGINEERING\SE LAB FOLDER\L6_L7_05_blood_donation_screening\Lab 6\.ven
v\Scripts\python.exe
cachedir: .pytest_cache
rootdir: C:\Users\Amoghaa R\Desktop\ENGINEERING 5TH SEMESTER\SOFTWARE ENGINEERING\SE LAB FOLDER\L6_L7_05_blood_donation_screening\Lab 6
collected 18 items

BranchCoverageTest_PES2UG24CS625_application.py::test_invalid_cohort_type PASSED [  5%]
BranchCoverageTest_PES2UG24CS625_application.py::test_invalid_volume PASSED     [ 11%]
BranchCoverageTest_PES2UG24CS625_application.py::test_valid_empty_cohort_list PASSED [ 16%]
BranchCoverageTest_PES2UG24CS625_application.py::test_unregistered_cohort PASSED [ 22%]
BranchCoverageTest_PES2UG24CS625_application.py::test_registered_cohort_without_donors PASSED [ 27%]
BranchCoverageTest_PES2UG24CS625_application.py::test_donor_below_minimum_age PASSED [ 33%]
BranchCoverageTest_PES2UG24CS625_application.py::test_donor_below_minimum_weight PASSED [ 38%]
BranchCoverageTest_PES2UG24CS625_application.py::test_donor_with_low_hemoglobin_standard_cohort PASSED [ 44%]
BranchCoverageTest_PES2UG24CS625_application.py::test_low_hemoglobin_high_altitude_cohort PASSED [ 50%]
BranchCoverageTest_PES2UG24CS625_application.py::test_high_hemoglobin_donor PASSED [ 55%]
BranchCoverageTest_PES2UG24CS625_application.py::test_accepted_donor_with_fast_track PASSED [ 61%]
BranchCoverageTest_PES2UG24CS625_application.py::test_high_deferral_with_zero_yield PASSED [ 66%]
BranchCoverageTest_PES2UG24CS625_application.py::test_high_deferral_with_positive_yield PASSED [ 72%]
BranchCoverageTest_PES2UG24CS625_application.py::test_no_centrifuge_data PASSED [ 77%]
BranchCoverageTest_PES2UG24CS625_application.py::test_centrifuge_aborts_after_two_consecutive_low_readings PASSED [ 83%]
BranchCoverageTest_PES2UG24CS625_application.py::test_centrifuge_recovers_after_low_reading PASSED [ 88%]
BranchCoverageTest_PES2UG24CS625_application.py::test_centrifuge_aborts_after_recovery_and_two_low_readings PASSED [ 94%]
BranchCoverageTest_PES2UG24CS625_application.py::test_centrifuge_multiple_stable_runs PASSED [100%]

============================== 18 passed in 0.06s ==============================
(.venv)
Amoghaa R@DESKTOP-F7VHOTL UCRT64 /c/Users/Amoghaa R/Desktop/ENGINEERING 5TH SEMESTER/SOFTWARE ENGINEERING/SE LAB FOLDER/L6_L7_05_blood_donation_screening/Lab 6
$ python -m coverage report -m
Name                          Stmts   Miss Branch BrPart  Cover   Missing
-----------------------------------------------------------------------
blood_donation_screening.py      45      0     32      0   100%
-----------------------------------------------------------------------
TOTAL                            45      0     32      0   100%
(.venv)
```